



How Push Notifications Actually Work

From Beginner to Production Systems

1. The Question

How does Zomato send notifications to 50 Million users?

Most beginners think:

```
for(User user : users){  
    sendNotification(user);  
}
```

Looks correct.

But imagine

```
Users = 50 Million
```

That means

```
50 Million Requests
```

This will create problems:

- Huge latency
- Network overload
- API limits
- Memory pressure
- Retry storms
- Cost explosion

Production systems don't work like this.

2. Real Production Flow

Marketing Team



Campaign Service



Notification Event



Kafka / RabbitMQ / SQS



Worker Pool



Firebase Cloud Messaging



Google Push Servers



Google Play Services



Android OS



Notification

Every box has one responsibility.

3. Why Queue?

Imagine

Marketing sends

50 Million Notifications

Instead of sending immediately,

store them inside a Queue.

Why?

Because Queue provides

- Buffer
- Retry
- Scalability
- Backpressure handling
- Parallel processing

Without Queue

↓

Server crashes.

4. Why Kafka?

Kafka is **optional**.

Companies may use

- Kafka
- RabbitMQ

- AWS SQS
- Azure Service Bus
- Google Pub/Sub

Purpose is always same

Store

↓

Distribute

↓

Scale

5. Worker Pool

Queue doesn't send notifications.

Workers do.

Example

Worker 1

Worker 2

Worker 3

...

Worker 500

Each Worker

Pull

↓

Process



Send to Firebase

More workers



Higher throughput.

6. Why Firebase?

Can Zomato directly notify your phone?

No.

Because Zomato doesn't know

- Is phone online?
- Is app alive?
- Network?
- Battery restrictions?
- Android version?

Firebase solves all these.

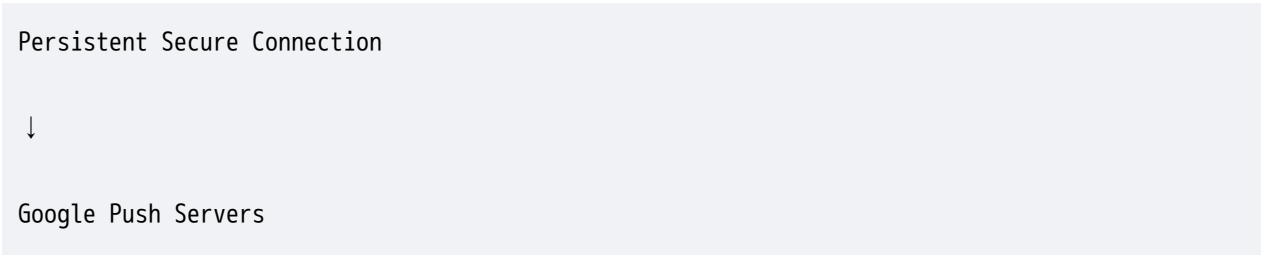
7. Google Push Infrastructure

This is the real magic.

Every Android phone already runs

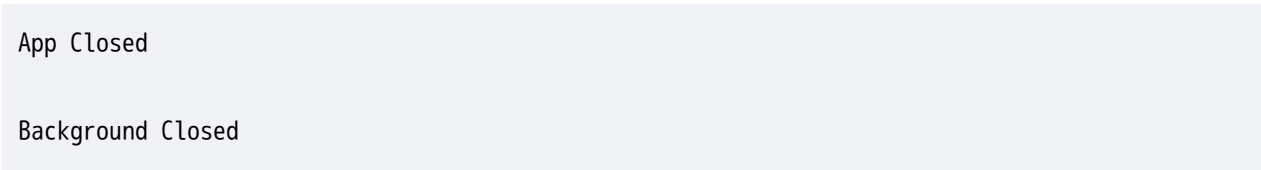
Google Play Services

It maintains



24×7

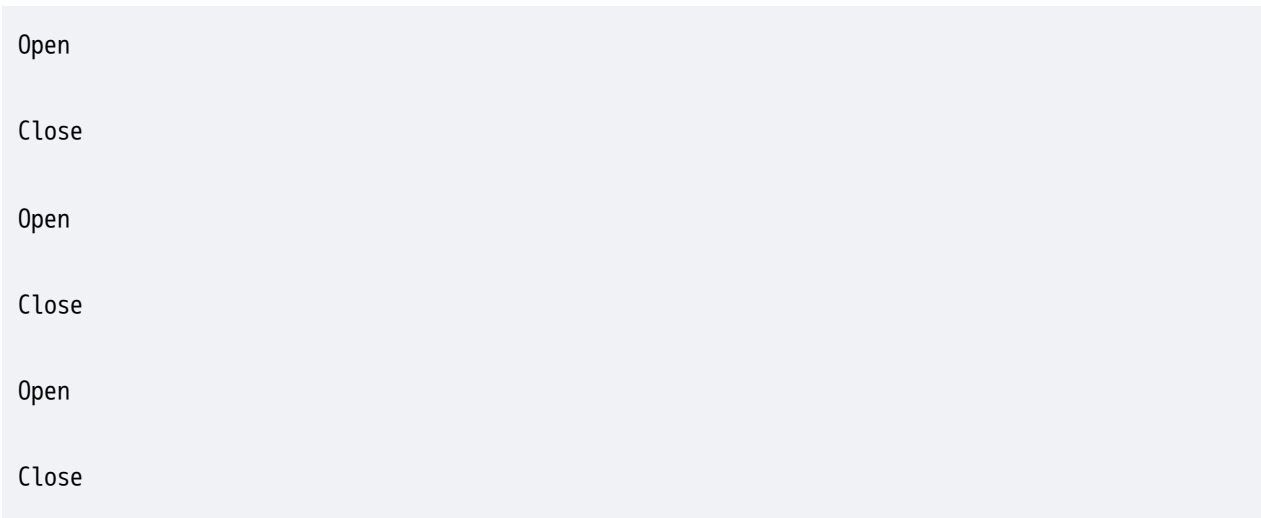
Even if



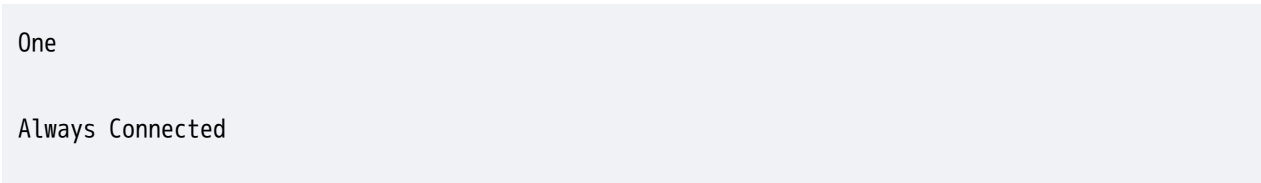
Connection is still alive.

8. Persistent Connection

Instead of



Google keeps



Channel

Benefits

- Faster delivery
- Less battery usage
- Less CPU
- Less network overhead

9. Device Token

Every app installation receives

Device Token

Like

ABC123XYZ...

This is **not**

User ID.

This is

Phone Address.

Zomato stores

User

↓

Device Token

When notification comes

↓

Firebase receives

↓

Device Token

↓

Find phone

↓

Deliver.

10. Apartment Analogy

Zomato

↓

Courier

↓

Security Guard

↓

Flat 702

Google Play Services

↓

Security Guard

Your Phone

↓

Flat

Zomato never goes to every flat.

11. Why Battery Doesn't Drain?

Imagine

100 Apps

Each creates

Own WebSocket

Battery 💀

Instead

Google says

Everyone use my connection.

Only one connection.

Huge battery savings.

12. Does Firebase send to 50 Million Phones?

Yes.

Eventually

Every phone receives

Individual notification.

Difference is

Before

Zomato

↓

50 Million Deliveries

Now

Zomato

↓

Firebase

↓

Google Push Servers

↓

50 Million Phones

Fan-out happens inside Google's infrastructure.

13. Fan-Out

Very important interview concept.

One message

↓

Millions of receivers.

This is called

Fan-Out

Google handles this.

Not Zomato.

14. Backpressure

Producer

100K msg/sec

Consumer

10K msg/sec

Problem

↓

Queue grows.

Solutions

- Buffer
 - More Workers
 - Rate Limit
 - Retry
 - DLQ
 - Autoscaling
-

15. Why Production Doesn't Use for-loop?

Because production needs

- Retry
- Parallelism
- Monitoring
- Scaling
- Fault tolerance
- Delivery guarantees

A simple loop gives none of these.

16. Interview Questions

Q1

Why Queue?

Q2

Why Worker Pool?

Q3

What if Firebase is down?

Q4

How would you retry failed notifications?

Q5

How to scale from

1 Million

↓

100 Million

users?

Q6

Difference between

Push

Pull

Polling

Q7

Explain Backpressure.

Q8

What is Fan-Out?

Q9

Persistent Connection vs HTTP Request

Q10

Why doesn't Android keep one WebSocket per app?